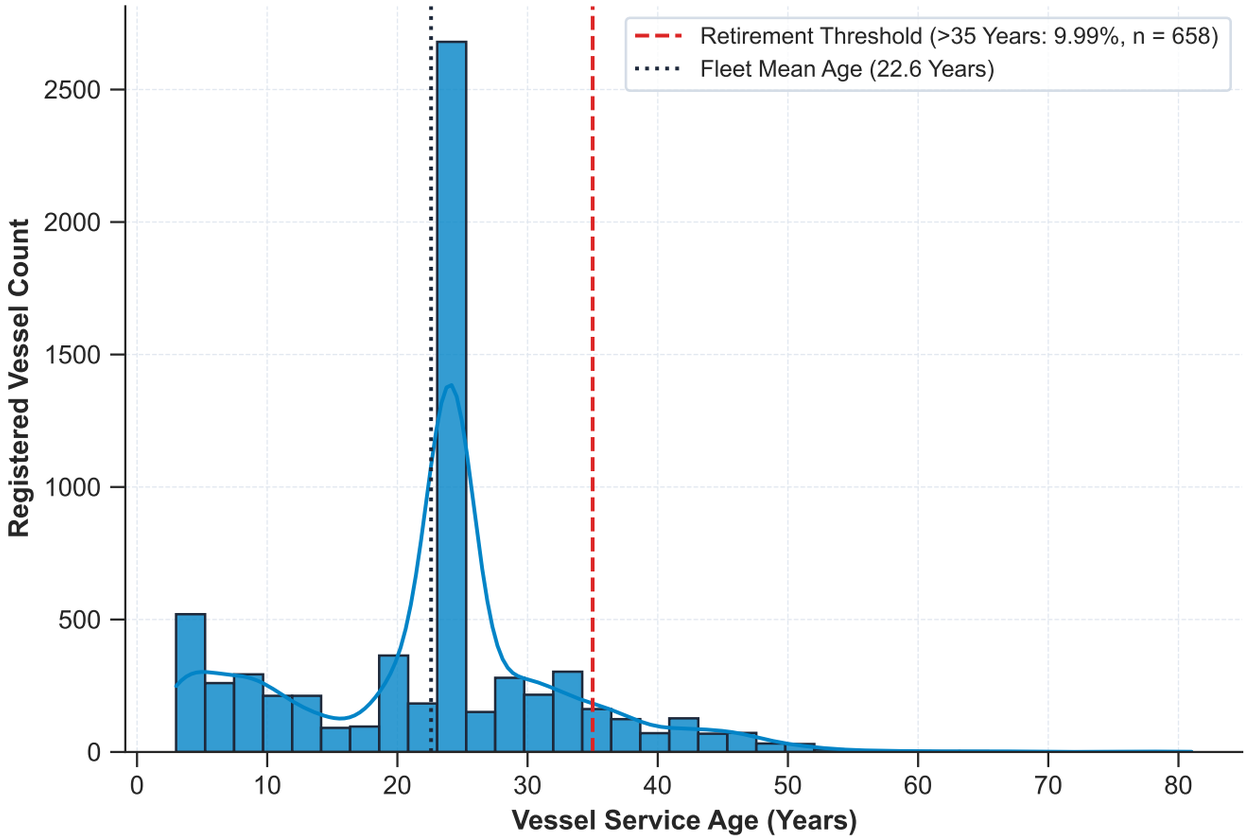
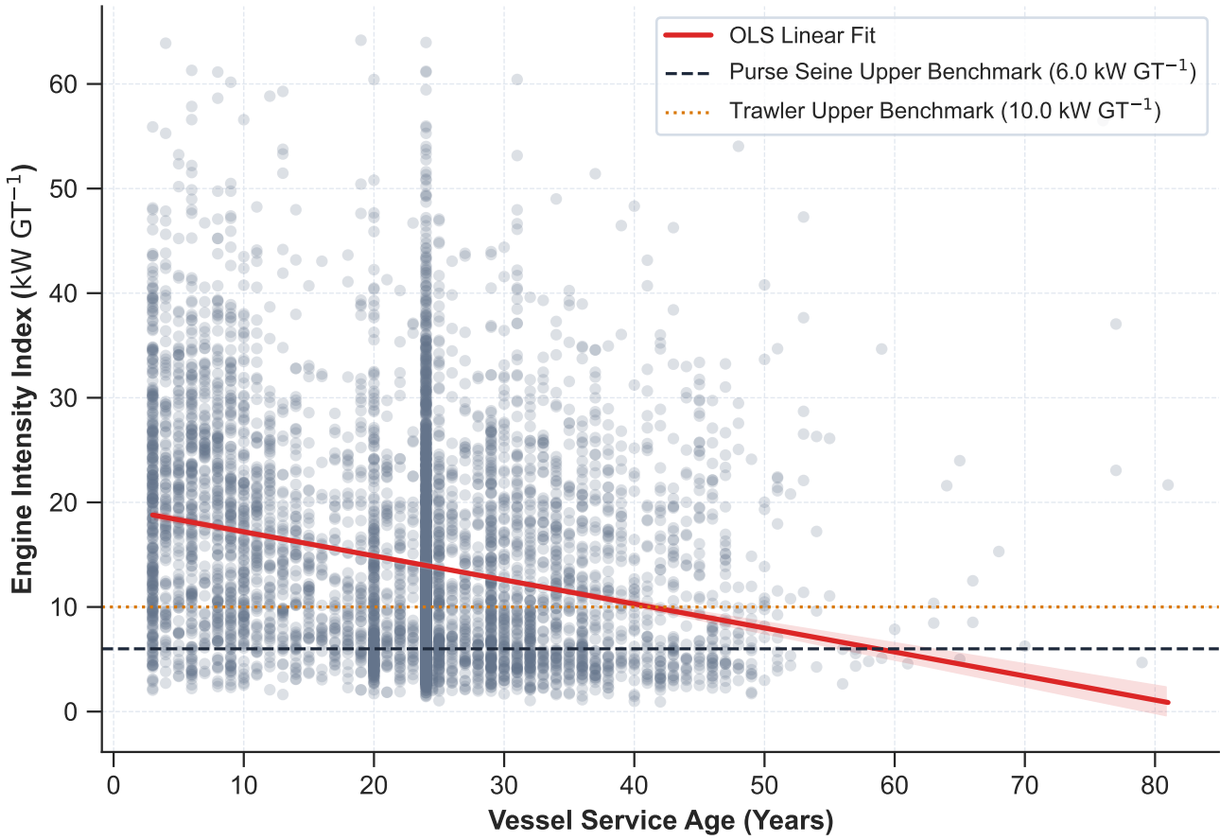
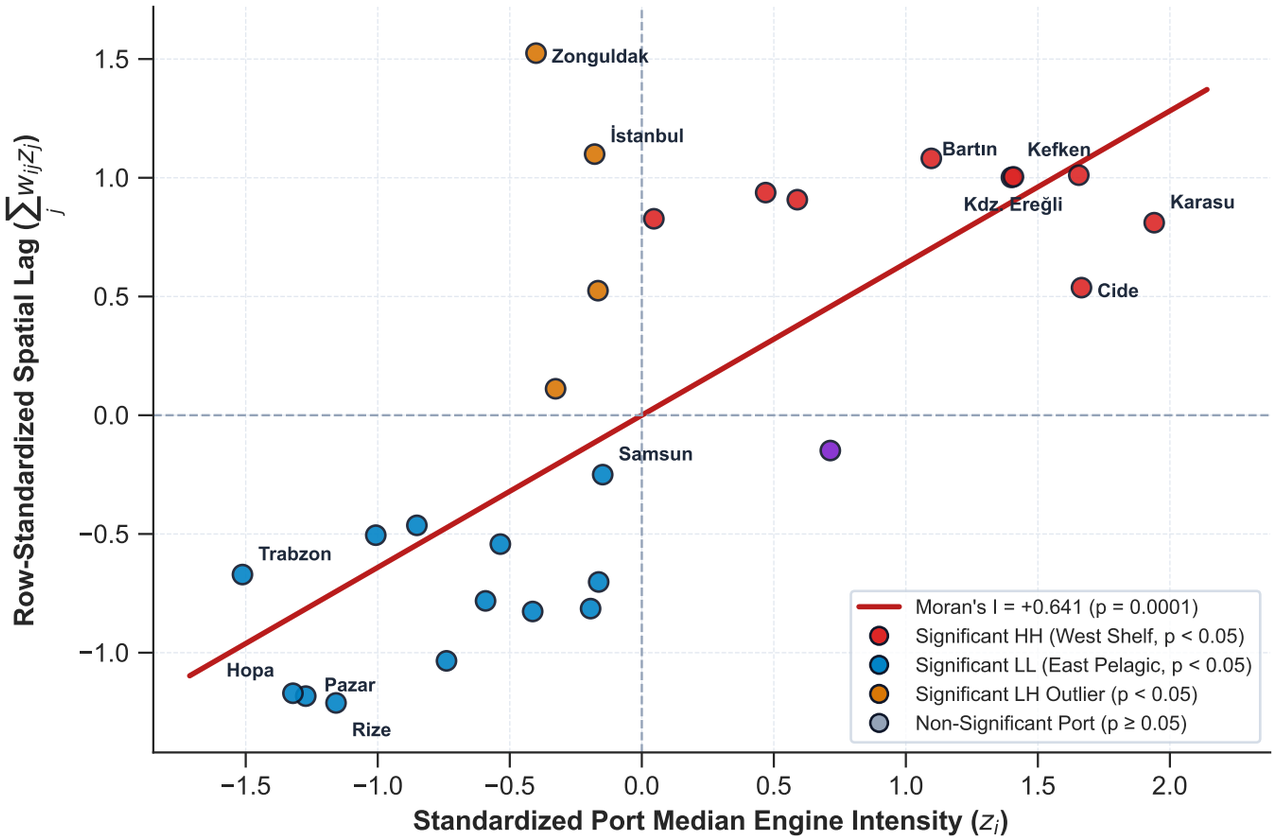
(a) Age Distribution of Registered Turkish GSA 29 Vessels (N = 6,584)



(b) Vessel Age vs. Engine Intensity Across Registered Vessels



(c) Empirical Moran Scatter Plot: 27 Black Sea Ports ($k = 4$ Nearest)



(d) Parametric CO_2 Sensitivity: Duty-Cycle Baseline vs. Uniform Bounds

